

Outsourcing Artificial Intelligence: Responding to the Reassertion of the Human Element into Automation

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ABSTRACT

Science fiction imaginaries and Silicon Valley innovators have long envisioned a workerless future. However, this industrial ambition has often outpaced technological realities in robotics and artificial intelligence, leading to a reassertion of human skills to cover untenable gaps in autonomous systems. This one-day workshop will invite discussion on this recent retrograde trend toward precarious (and often concealed) human labour across such domains as agriculture, transportation, and caregiving through paper presentations and design activities. Throughout, we will ask how this phenomenon speaks to engineering and design challenges and, subsequently, encourage participants to ideate new cybernetic arrangements that centre the agency and well-being of essential workers.

CCS CONCEPTS

• **Social and professional topics** → Professional topics; Computing and business; Automation; • **Human-centered computing** → Interaction design; Interaction design process and methods; User centered design; • **Applied computing** → Law, social and behavioral sciences; Anthropology; Ethnography.

KEYWORDS

Human-computer interaction, artificial intelligence, robotics, essential labor

ACM Reference Format:

Mathew Iantorno, Olivia Doggett, Priyank Chandra, Julie Yujie Chen, Rosemary Steup, Noopur Raval, Vera Khovanskaya, Laura Lam, Anubha Singh, Sarah Rotz, and Matt Ratto. 2022. Outsourcing Artificial Intelligence: Responding to the Reassertion of the Human Element into Automation. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3491101.3503720>

1 BACKGROUND

From the introduction of the robotic arm on the factory floor to modern data-driven agricultural startups, industry has long sought to remove the human element from labour. Yet despite the intensified development of robotics and artificial intelligence by Silicon Valley companies, the technological realities of today have prevented us from achieving the workerless society promised in the fictions of tomorrow. In many regards, we have moved in retrograde: self-driving cars have been curbed in favour of gig economy workers [1]; content moderation on Facebook is often entrusted to call centres rather than algorithms [2]; and delivery robots on North American college campuses are being teleoperated from South America [3]. Although this reassertion of the human element is often framed as a stopgap measure, it brings into question common assumptions regarding which embodied and intuitive skills can easily be subsumed by machine learning.

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CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

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ACM ISBN 978-1-4503-9156-6/22/04.

<https://doi.org/10.1145/3491101.3503720>

This virtual workshop will address the rise of this form of outsourcing artificial intelligence across such major sites of labour as agriculture, transportation, and caregiving. It will build on previous critical scholarship on robotics, artificial intelligence, and the future of work [4–6]; political economy pieces on precarious labour and the gig economy [7, 8]; ethnographic studies of essential workers and their interactions with technology across various sites of labour [1, 9–11]; policy reports on the challenges and ethics of artificial intelligence [12, 13]; and methodologies for critical and collaborative design thinking [14–16].

Throughout, questions such as the following will be posed: To what extent does the historical devaluation of essential labour (and those who perform it) play in deciding which abilities are amenable to automation? How have science fiction imaginaries and Silicon Valley visionaries alike subsequently appraised and shaped the physical and mental demands of labour? How have new forms of precarious labour positioned and created new demands for the human body and human proficiencies? And, lastly, how does this reassertion of the human element into autonomous systems speak to engineering and design challenges? Through these inquiries, we aim to move past broad criticisms of capital and towards constructive design thinking regarding the future of automation.

This workshop has three overarching objectives. First, we aim to build a scholarly community that fills an interdisciplinary gap in the study of labour and automation. We will do so by hosting a collaborative conversation between two adjacent fields: computer science and human-computer interaction (HCI) scholars, who are tasked with developing and redeveloping technologies that emulate human skills; and social science and science and technology studies (STS) scholars, who attend to the historical context of automation and the modern sites of labour where it has taken hold.

Creating this community will allow us to pursue our second objective: defining the unique research and design challenges presented by the reassertion of human skill into these technological assemblages. By inviting new scholarship on automation in the spheres of agriculture (such as data-driven farming), transportation (such as autonomous vehicles), caregiving (such as assistive robotics), and beyond, we will position the phenomenon of outsourcing artificial intelligence as a distinct concern within the field of HCI that requires new methods of scrutiny. We will move past dismissing such challenges as ephemeral technical shortcomings and recontextualize them within longstanding misappraisals of human and machinic ability – underlining how the ambition to replace marginalized labourers with technology often serves to reinforce their irreplaceability.

By engaging in this interdisciplinary and inter-industry analysis, we will pursue our third objective: ideating technologies that centre the agency and well-being of essential workers. In doing so, we will ask the following questions: Is it possible to create new equitable cybernetic relationships that embrace rather than erase the reassertion of the labourer? How can design interventions be practically developed and implemented alongside policy interventions, and vice versa? What kind of value alignments need to be developed to address these challenges across such labour domains as agriculture, transportation, and caregiving in a way that provides useful philosophical and design insights? And how does automation in these labour spheres unfold locally in different geographies?

2 ORGANIZERS

For this workshop, we have selected scholars who study the role of emerging autonomous technologies within existing labour practices, with a focus on the three highlighted domains. Our organizers are situated within STS, computer science, and/or HCI scholarship; attend to a variety of methodological, theoretical, and critical approaches; and have engaged in a diversity of case studies across the Global North and Global South.

Priyank Chandra is an assistant professor in the Faculty of Information at the University of Toronto. His research studies the sociotechnical practices of communities living at the margins of society. Specifically, he focuses on how informality shapes technology use, sustains collaboration, and creates resilient practices. He received his PhD in Information from University of Michigan.

Julie Yujie Chen is an assistant professor at the Institute of Communication, Culture, Information, and Technology (ICCIT) at the University of Toronto Mississauga with a graduate appointment at the Faculty of Information. Chen studies the transformation of work and the political economy of media and communication in relation to digital technologies, capitalism, and globalization, particularly concerning the issues of marginalization and worker agency.

Olivia Doggett is a doctoral student in the Faculty of Information at the University of Toronto. Drawing from the disciplines of critical computing, design theory, information and environmental studies, her research explores the role of information services and technologies on agricultural production and farm labour.

Mathew Iantorno is a doctoral student in the Faculty of Information at the University of Toronto. His research focuses on conceptualizing accessible and just human-robot interactions, specifically regarding automata deployed in caregiving and other day-to-day scenarios. He currently holds a graduate fellowship with the Ethics of AI Lab within the Centre for Ethics at the University of Toronto.

Vera Khovanskaya is a postdoctoral research fellow in the Communication Department at the University of California, San Diego. She studies how data collection in the workplace shapes the experience of work and creates opportunities for advocacy. Her recent work has examined how worker advocates have used workplace data in the past; how tactics used by labor unions to strategically engage with industrial scientific management in the mid-20th century can inform data-driven worker advocacy in platform-mediated work; and how contemporary data-collection technologies impact union organizing.

Laura Lam is a doctoral student at the Centre for Industrial Relations and Human Resources at the University of Toronto and researcher with the Canada Excellence Research Chair (CERC) in Migration and Integration. Her research interest is at the nexus of migration, precarious employment, and gender, with a focus on the use of app-based digital labour platforms.

Matt Ratto is an Associate Professor in the Faculty of Information at the University of Toronto and the Bell University Labs Chair in Human-Computer Interaction. His research focuses on how theories and perspectives from technoscience research can usefully extend and contextualize design and engineering practice, particularly related to emerging digital health technologies. He studies and practices “critical making,” work that combines humanities insights

and engineering practices, and has published extensively on this concept.

Noopur Raval is a postdoctoral research fellow at the AI Now Institute at New York University. She received her PhD in Informatics from the University of California Irvine in September 2020. Her dissertation, entitled *Platform Living – theorizing life, work and ethical enactments in the gig economy*, looked at work, consumption, and life experiences within gig platforms through ethnographic research in Bengaluru (among other places) and the moral economy of the gig economy.

Sarah Rotz is an assistant professor in the Faculty of Environmental and Urban Change (FEUC) at York University. Her academic and organizing work is grounded in environmental justice, with a focus on land and food systems. Her work has focused on topics ranging from the political economy and ecology of farmland tenure and critical perspectives of big data in agriculture, to the ways that settler colonial logics and gendered narratives uphold extractive practices and relationships on the land.

Anubha Singh is a doctoral student in the School of Information at the University of Michigan. Through ethnographic inquiry, she studies the integration of technology in agribusinesses in India, especially relating to food production and distribution in long food supply chains.

Rosemary Steup is a doctoral student in the Computing, Culture, and Society track of the Informatics PhD program at Indiana University. Her current research involves the design, marketing, and adoption of digital farming technologies. She has a B.S. in Computer Science and French from Trinity University.

3 WEBSITE

We have developed a website dedicated to the promotion of and recruitment for this workshop at <https://criticalautomation.org/outsourcing-ai/>. This is a standard WordPress site built to WCAG accessibility standards. Moving forward, it will provide basic information and promotional materials for the workshop as well as a Google Form for the submission of position papers. Additional applications of the website will be noted throughout the Pre-Workshop Plans and Post-Workshop Plans sections.

4 PRE-WORKSHOP PLANS

Recruitment and community-building will occur through both our website and institutional networking. The website will serve as the primary promotional hub, presenting the formal call for participation as well as broader information regarding the workshop structure and organizers. We intend to forward the website URL alongside the formal call for participation to affiliated institutions and faculties to circulate through their own communication channels. This will include the McLuhan Centre for Culture and Technology; the Labor Tech Research Network; the Design Justice Network; and Ryerson University's Migration and Integration Canada Excellence Research Chairs (CERC) program. The use of these interdisciplinary and inter-university networks will ensure a diversity of perspectives within the workshop.

Additional recruitment will take place through the ongoing activities of the co-organizer's working group, *We Can Rebuild You:*

The Automation and Erasure of Essential Workers. Hosted at the University of Toronto's McLuhan Centre for Culture and Technology but engaged virtually, this inter-institutional working group brings together HCI and humanities scholars to discuss labour and technology. Pertinently, we will be engaging in a speaker series prior to CHI 2022, partly in the aim of soliciting interest in the workshop. These lectures, intended to be virtual and public, will pair scholars with industry contacts and labour organizers to discuss the move toward automation within the aforementioned spheres of labour. These lectures will be circulated through the various communication channels used for recruitment.

The call for participation will be presented as a page with a linked Google Form on our website. We will be requesting formal position papers that are 2-4 pages long and adhere to general ACM formatting and CHI standards. Priority will be given to papers that clearly attend to the central thesis of the workshop; speak explicitly to a labour sphere such as agriculture, transportation, or caregiving; and discuss prescient technologies and sites of labour across various geographies and cultures. These position papers will be selected by the conference organizers through independent review and virtual adjudication over Zoom.

Once participants have been chosen, we will take a number of steps to ensure the workshop runs smoothly and is inclusive of all participant needs. First, we will host preparatory Zoom meetings to sort through organizational concerns and confirm our participant list. Second, we will distribute a survey through our mailing list that acknowledges accessibility needs, pronoun use, and time zone differences for both our own benefit and that of the conference organizers. Third, we will post all necessary materials to the workshop website. This will include a finalized agenda of workshop activities; a subsequent agenda for post-workshop activities; a shared repository of all position papers; short biographies of all participants; and an abridged reading list of noteworthy citations. Fourth, we will continue promotion of the workshop through the aforementioned online networks.

5 WORKSHOP STRUCTURE (AND REMOTE/ONSITE PLANS)

This one-day workshop will be a wholly online event that takes place over six working hours. Workshop organizers will work alongside 15-20 participants in discussion and design activities regarding automation, labour, and the specific imperative of outsourcing artificial intelligence. The workshop will be divided into a morning and an afternoon session with a lunch break afforded in between.

The morning session will consist of position paper presentations and general discussion. Presentations will be 5 minutes in length and divided into 4 blocks of 3-5 papers based on theme and/or domain of labour. Each block will be followed by a question-and-answer period. Workshop presentations will take place over Zoom, and all presentations will be auto-captioned and recorded both for accessibility purposes and to enable asynchronous engagement. Attendees will also be provided the option of pre-recording their full presentations ahead of time.

The afternoon session will consist of collaborative design activities. Participants will be divided into three Zoom break out rooms. Each group will discuss the unique design challenges provoking the

reassertion of the human element into automation in their domain and then ideate new solutions that are inclusive of technological realities and worker agencies. This ideation will take the form collaborative brainstorming, drawing, and prototyping using the Miro platform. A larger group discussion will follow to find alignments in these solutions across labour domains.

5.1 Workshop Schedule

- 9:00 – 9:20 Welcome and introductions
- 9:20 – 9:40 Workshop overview and morning session agenda
- 9:40 – 10:10 Presentation block 1
- 10:10 – 10:40 Presentation block 2
- 10:40 – 10:50 *Coffee break*
- 10:50 – 11:20 Presentation block 3
- 11:20 – 11:50 Presentation block 4
- 11:50 – 12:00 Wrap up and summary
- 12:00 – 1:00 *Lunch break*
- 1:00 – 1:10 Afternoon session agenda
- 1:10 – 2:40 Breakout design activities
- 2:40 – 2:50 *Coffee break*
- 2:50 – 3:50 Group discussion
- 3:50 – 4:00 Wrap up and future work

**We are open to scheduling one or two early morning or evening slots to accommodate participants in different time zones.*

6 POST-WORKSHOP PLANS

The principal output of this workshop will be a special issue of a journal. Participants in the workshop will be encouraged to refine, expand upon, and re-submit their position papers as full-length conference proceedings based on discussions carried out throughout the workshop. There will also be a subsequent call-for-papers following the conference in early 2022 to fill in any perceived gaps in scholarship. In both instances, submissions will be due October of 2022. These papers will be subject to a rigorous peer review process, handled virtually by workshop organizers. This special issue will be pitched to an open access journal with the mandate of encouraging widespread access to this prescient topic of study.

Following the conference, we will also publish a workshop report on our website. This report will serve as an abridged summary of the workshop and will acknowledge addressed and unaddressed research questions, setting the stage for future work. In pursuit of this future scholarship, we intend on establishing a *Critical Automation Studies* mailing list, inviting scholars to share their research of this topic beyond the confines of the conference and our associated working group.

7 CALL FOR PARTICIPATION

Science fiction imaginaries and Silicon Valley innovators have long envisioned a workerless future. However, this industrial ambition has often outpaced technological realities in robotics and AI, leading to a reassertion of human skills to cover gaps in autonomous systems. By examining trends toward precarious labour across such domains as agriculture, transportation, and caregiving, this workshop will ask how this phenomenon speaks to engineering and design challenges. In doing so, we aim to ideate new cybernetic

arrangements that centre the agency and well-being of essential workers.

We are accepting position papers 2-4 pages in length related to this theme. Submissions should attend to the central thesis of the workshop; speak explicitly to a labour sphere and issues of automation; address how such challenges unfold locally in various geographies and cultures; and adhere to general ACM formatting. We welcome auto-ethnographic work from those with direct experience with autonomous labour systems.

This is a two-part online workshop that will take place over one six-hour session. The morning session will consist of position paper presentations and general discussion; the afternoon session will see participants divided into breakout groups to discuss the unique research and design challenges of outsourcing artificial intelligence. At least one author must be able to attend the workshop virtually, and all participants must register for at least one day of the conference.

Information regarding submission and pre- and post-workshop activities will be made available through our website: <https://criticalautomation.org/outsourcing-ai/>. Accepted papers will be made available online prior to the workshop.

ACKNOWLEDGMENTS

The authors would like to thank the McLuhan Centre for Culture and Technology for supporting the development of this workshop and the preceding working group.

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